

Cross-Chain Exploit Propagation and Systemic Risk Modeling

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Abstract

Cross-chain bridges and interoperable DeFi protocols introduce complex contagion risks where localized exploits cascade across ecosystems, threatening systemic stability. This project develops a formal multi-layer temporal graph model of cross-chain ecosystems to simulate exploit propagation, estimate exposure coefficients, and compute systemic risk metrics such as conditional expected shortfall on total value locked. Extending prior team research on security analysis [3, 6, 7, 2, 4, 5], the framework enables stress-testing and evaluation of interventions like bridge throttling. Recent literature on liquidity exhaustion and aggregated risk indices [8, 9] highlights the urgency. The project will release an open-source simulator and risk API.

Problem Formalization

Existing cross-chain risk models are limited by static or single-chain assumptions that fail to capture dynamic inter-layer liquidity flows and temporal contagion, leading to underestimation of systemic impact from bridge exploits. This project solves these by constructing a multi-layer temporal graph with Bayesian parameter inference on real bridge/oracle data and agent-based adversarial simulations, producing quantifiable CES risk scores and intervention efficacy analysis.

The proliferation of cross-chain bridges and composable DeFi constructs creates pathways for contagion: breaches on one chain rapidly affect liquidity and solvency elsewhere. This project constructs a formal, data-driven model of cross-chain exploit propagation and quantifies systemic risk measures that help stakeholders prioritize hardening and insurance.

We model a multi-chain ecosystem as a multi-layer temporal graph $\mathcal{G} = \{G_t^{(c)}\}_{c \in \mathcal{C}, t \in \mathbb{T}}$ where each layer $G_t^{(c)} = (V_t^{(c)}, E_t^{(c)})$ corresponds to chain c at time t , nodes represent protocols/accounts, and inter-layer edges model cross-chain flows via bridges, wrapped assets and messaging. Liquidity position of node v on chain c at time t is $L_v^{(c)}(t)$. An exploit begins as a localized shock ξ that reduces liquidity of a subset $S \subset V^{(c)}$; propagation occurs via asset re-anchoring, price oracles, and liquidation cascades. A discrete-time contagion model writes:

$$L_v(t+1) = L_v(t) - \sum_u \kappa_{u \rightarrow v}(t) \cdot \Delta L_u(t) - \eta_v(t),$$

where $\kappa_{u \rightarrow v}(t)$ captures cross-asset exposure (a temporally-dependent coupling), and $\eta_v(t)$ models endogenous shocks (e.g., slippage, MEV extraction). Systemic risk is quantified by tail metrics such as conditional expected shortfall (CES) on total locked value across all layers:

$$\text{CES}_\alpha = \mathbb{E} \left[\sum_{c,v} \max(0, -\Delta L_v^{(c)}) \mid \sum_{c,v} \max(0, -\Delta L_v^{(c)}) \geq q_\alpha \right],$$

with threshold q_α being the α -quantile.

Empirically estimating κ and η requires historical cross-chain transaction reconstruction, bridge event logs, wrapped-asset issuance/retire events, and oracle feed histories. We adopt a Bayesian inference approach to fit parameters with uncertainty, enabling stress-testing under plausible but rare scenarios (e.g., simultaneous oracle manipulation plus bridge re-entrancy). The project also explores control interventions (e.g., dynamic bridge throttling, insurance-backed circuit breakers) and computes their efficacy by solving constrained optimization problems that minimize realized loss subject to usability constraints.

Methodologically, the project blends temporal multilayer graph learning (to predict propagation pathways), agent-based simulation (to model strategic adversaries and MEV actors), and risk-theoretic analysis. A key novelty is the integration of economic models (liquidity curves, slippage functions) with graph contagion, producing actionable risk scores for protocols and assets. Outputs include an open-source simulator, a cross-chain exposure dataset, and a risk-scoring API that nodes and custody services can query.

We seek collaborations with bridge operators, oracles, and DeFi protocol teams to obtain higher-fidelity data and to co-design mitigations. We also invite financial engineers interested in designing parametric insurance products and regulators or self-regulatory organizations concerned with cross-chain systemic stability.

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